Hyperspectral Image Super-Resolution Using Differentiation and Cross-Domain Feature

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***Abstract*—The rich spectral information in hyperspectral images (HSIs) require effective joint extraction of spectral and spatial features for super-resolution (SR) tasks. To better capture spectral–spatial representations and restore edge textures, we propose the differentiation and cross-domain feature extraction network (DCDFENet) for HSI SR. Its core module enhances reconstruction by leveraging cross-domain interactions and differential features. Differentiation and cross-domain feature extraction (DCDFE) consists of three components: cross-domain and multiscale feature extraction (CDMSFE), which uses separated 3-D convolutions and cross-domain connections to extract complementary features at multiple scales; spectral feature differentiation highlighting attention (SFDHA), which emphasizes significant spectral variations via differentiation and attention mechanisms; and edge feature extraction (EFE), which integrates Laplacian filtering and an extreme value feature extraction (EVFE) to enhance spatial edge textures. In addition, a loss function based on singular value decomposition (SVD) further mitigates spectral distortion. Experiments on three benchmark datasets demonstrate that our method achieves superior performance on peak signal-to-noise ratio (PSNR), mean PSNR (MPSNR), structural similarity (SSIM), and spectral angle mapping (SAM) metrics.**

***Index Terms*—Cross-domain feature extraction, extreme value feature, spectral differentiation, super-resolution (SR).**

I. Introduction

HYPERSPECTRAL images (HSIs) simultaneously capture high-dimensional spectral signatures and spatial details, offering comprehensive information about the observed scene. By segmenting the electromagnetic spectrum into hundreds of contiguous and narrow spectral bands, HSIs enable precise material identification and classification, making them valuable in diverse remote sensing applications such as land cover monitoring [1], [2], urban planning [3], road network extraction [4], and geological exploration [5].

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Despite their advantages, HSIs often suffer from low spatial resolution due to physical limitations of imaging sensors, which must balance spectral richness against spatial clarity under restricted energy and hardware budgets. This limitation has motivated the development of HSI super-resolution (HSI SR) techniques to enhance spatial details while maintaining spectral integrity. Existing HSI SR methods are mainly divided into two categories: traditional approaches that incorporate handcrafted priors within optimization frameworks and deep learning-based methods that learn nonlinear mappings from low-to-high-resolution (HR) representations.

Traditional algorithm-based reconstruction methods leverage prior image knowledge to constrain the solution space and reduce the complexity of reconstruction. Early efforts in HSI SR reconstruction focused primarily on various mathematical techniques, such as interpolation [6], [7], sparse coding [8], [9], and wavelet transform [10], [11]. Despite their simplicity, these methods often produce blurred reconstructions and yield suboptimal results.

Deep learning, recognized for its efficacy in numerous computer vision tasks, has been applied to SR. Convolutional neural networks have become the main tool in this area. The introduction of a convolutional neural network for the image SR task was first realized by Yoon et al. [12], marking the transition from traditional algorithms to deep learning. To achieve SR of real-world remote sensing images with different qualities in a unified framework, Dong et al. [13] proposed a practical degradation model and a kernel-aware network (KANet). While deep learning-based SR technology for natural images has achieved a relatively advanced level [14], [15], [16], [17], methods tailored for HSIs are still quite scarce.

At present, deep learning-based HSI SR methods can be broadly classified into two main categories: fusion-based SR [18], [19], [20], [21], [22] and single-image SR (SISR). Fusion-based SR can be further categorized into fusion based on multispectral images (MSIs) [23], [24], [25], [26] and fusion enhanced by panchromatic (PAN) sharpening [27], [28], [29]. Fusion-based methods rely on additional data to inject spatial details. Palsson et al. [30] combined principal component analysis (PCA) with 3-D convolutional networks to address the MSI-HSI fusion problem as early as 2017. In 2022, Hu et al. [31] proposed a transformer-based fusion network (Fusformer) to exploit long-range spatial–spectral dependencies. PAN images typically exhibit superior spatial resolution. Qu et al. [32] developed a multiresolution spatial–spectral feature learning (MSSL) framework for the fusion of hyper-

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spectral and PAN images. In addition, employing multiple shallow networks for feature extraction can significantly reduce computational costs. However, fusion-based HSI SR methods still encounter limitations, such as challenges in image registration and alignment, poor generalization capabilities, and in terms of the difficulty in data acquisition.

SISR approaches reconstruct HR HSIs from low-resolution counterparts without requiring auxiliary modalities. Its intrinsic flexibility and applicability have attracted considerable attention in practical applications. Yuan et al. [33] implemented deep learning to adapt a convolutional neural network, originally trained on natural images, for HSIs processing. Hu et al. [34] proposed a deep spectral differential CNN (SDCNN) that incorporates a spatial error correction model, with the goal of enhancing spatial detail while preserving spectral integrity. Guo et al. [35] utilized a dense structure that integrates both nonlocal and local blocks, effectively combining local and global features to reconstruct HSIs with improved fine details. Dong et al. [36] introduced a context-aware guided attention-based cross-feedback dense network that utilizes a feedback mechanism between two branches, offering robust spatial–spectral guidance for low-level feature extraction. Early methods based on 2-D convolutions focused solely on spatial information while neglecting the rich spectral content.

To extract the spectral dimension information from HSIs, numerous methods for HSI SR utilizing 3-D convolution have been proposed. Mei et al. [37] proposed a 3-D fully convolutional network (3DFCNN) that utilizes 3-D convolutions to directly capture spatial context from neighboring pixels. This approach efficiently extracts spectral correlations across adjacent bands, which helps to minimize spectral distortion in the reconstructed image. However, 3-D convolutions inevitably bring in a great number of parameters. To mitigate the rise in network parameters and computational complexity caused by 3-D convolution, Huynh et al. [38] developed a modulation classification network (MCNet) that utilizes 2-D/3-D separable convolutions to reduce computational complexity.

Meanwhile, attention mechanisms have also been widely incorporated into networks to extract salient information. Li et al. [39] introduced an HSI SR technique that incorporates a band attention mechanism within an adversarial learning framework. Liu et al. [40] proposed spectral grouping and attention-driven residual dense network (SGARDN), which aims to model all spectral bands effectively while placing emphasis on the extraction of spatial–spectral features. And inspired by the self-attention mechanism, Ma et al. [41] introduced a network that combines a 3-D convolutional neural network with a transformer prior. This integration effectively captures spatial and spectral correlations inherent in HSIs, addressing the limitations associated with the learning capacity of traditional CNNs. However, the computational complexity of the Transformer is significantly higher than that of CNNs, leading to an increase in computational costs.

However, existing super-resolution (SR) networks still face limitations in jointly extracting spatial and spectral features, with insufficient use of the rich interband variations inherent in HSIs. Mainstream spectral attention mechanisms typically focus on local correlations among adjacent bands, lacking explicit modeling of spectral discrepancies and long-range dependencies, thereby failing to fully exploit deep spectral information. Revealing and leveraging interband variation patterns to enhance feature discriminability remains a key challenge in hyperspectral reconstruction. Meanwhile, current methods exhibit limited sensitivity to edge textures, resulting in blurred contours and weakened details, which impair the performance of downstream tasks such as object recognition and segmentation. Therefore, there is a pressing need for a novel feature extraction strategy capable of simultaneously capturing spectral variation and spatial structural details to improve the overall reconstruction quality of HSIs.

Existing deep learning-based super-resolution networks have demonstrated promising performance; however, they still face several issues that affect their overall efficiency. These challenges can be summarized as follows:

1) Existing SR networks may exhibit insufficient integration between spatial and spectral features, limiting their ability to provide richer feature representations.
2) Existing spectral feature extraction mechanisms predominantly focus on local correlations among adjacent spectral bands, resulting in insufficient exploration and utilization of the global interspectral information.
3) Reconstructed images often suffer from blurred and missing edge details.

To solve these problems, we introduce the differentiation and cross-domain feature extraction (DCDFE) network for HSI SR reconstruction. The core of this network is the DCDFE module, which is designed to enhance reconstruction capabilities by leveraging the cross-domain relationships and differential features of HSIs and consists of three main components: cross-domain and multiscale feature extraction (CDMSFE), spectral feature differentiation highlighting attention (SFDHA), and edge feature extraction (EFE). Specifically, CDMSFE constructs complementary relationships between spectral and spatial domains through cross-domain connections and contains multiscale features. The SFDHA comprehensively extracts feature variation trends in the spectral dimension through spectral differentiation and implements an attention mechanism to regulate the range of spectral feature changes based on these differential features. Subsequently, EFE employs a Laplacian operator to extract edge texture features in the spatial domain and integrates an extreme value feature extraction (EVFE) module to mitigate noise effects on network performance. We posit that extreme features can, to a certain extent, reflect peripheral information as well. EVFE primarily utilizes a mask derived from the Hessian matrix of the second-order differential to filter extreme values from the first-order partial differential features. Finally, a singular value decomposition (SVD) loss function is introduced to address issues related to spectral distortion. Multiple experiments were conducted on three general datasets, demonstrating that the proposed DCDFE network outperformed existing hyperspectral super-resolution methods. The primary contributions of this study are as follows.

1) We propose CDMSFE, which constructs complementary relationships between spectral and spatial-domain 3-D

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convolution layers through cross-domain connections to achieve a composite multiscale feature extraction. This approach enhances feature representation quality while preserving fine-grained details.

2) We propose SFDHA, which captures spectral variation trends by applying spectral differentiation and incorporates an attention mechanism to adaptively modulate the importance of spectral changes. This facilitates more effective highlighting of key spectral features.
3) We propose EFE, which employs a Laplacian operator combined with an EVFE module to extract spatial edge texture features, thereby improving the preservation of spatial details.

II. Related Works

In the field of HSI SR, substantial progress has been made in improving network reconstruction capabilities. Building on insights from pioneering studies of SISR, researchers have investigated various approaches to address the challenges associated with the limited spatial resolution of HSIs.

A. Single-Image Super-Resolution

Early studies focused on improving feature extraction through novel convolutional modules. Inspired by a study of Xie et al. [42], who transformed a super-resolution task into a feature matrix decomposition problem (DFMF) to separate 3-D convolution models, Li et al. [43] introduced a hybrid convolution module that combines 2-D convolution and separable 3-D convolution to efficiently extract both spatial and spectral features, while minimizing the number of parameters and reducing the computational complexity. Li et al. [44] proposed an enhanced approach to explore the interaction between 2-D and 3-D convolutions. By sharing spatial information, model redundancy was reduced, leading to better reconstruction performance of the network. Another significant contribution was made by Jiang et al. [45], who proposed a deep neural network based on a spatial–spectral prior network (SSPSR) to fully exploit the spatial and spectral correlation information within HSIs. As convolution operations become more complex, the number of parameters and computational cost of the network continue to increase. Unsupervised learning strategies have gained traction to reduce reliance on labeled data. Li et al. [46] proposed an X-shaped interactive autoencoder network. By combining U-Nets and implementing cross-modality mutual learning, it effectively utilizes multimodal information, enhancing spatial–spectral features. A joint self-supervised loss facilitates unsupervised optimization, thereby reducing training costs. Li et al. [47] introduced a novel unsupervised model-guided approach for HSI SR, utilizing degradation models and multiscale attentional fusion to improve mapping learning, yielding superior results across various datasets. However, the performance of these methods still has certain limitations in practical applications.

B. Attention Mechanism

Attention mechanisms have emerged as a powerful tool for feature refinement. The EU2ADL network [48] enhances multispectral aided HSI super-resolution through the use of coupled autoencoders, a hybrid loss function, and attention-embedded degradation learning, which improves representation learning and minimizes distortions. In addition, Zhang et al. [49] developed a CNN-based super-resolution reconstruction method that employs multiscale feature extraction and a multilevel feature fusion structure, addressing the challenge of designing effective models for spectral segment feature learning in hyperspectral remote sensing images. Drawing inspiration from attention mechanisms and multiscale feature extraction in cloud detection works [50], [51], [52], Zhang et al. [53] proposed the spectral correlation and spatial high-low frequency information HSI super-resolution network (SCSFINet), utilizing spectrum-guided attention for analyzing information obtained from HSI, with a focus on addressing the challenge of obtaining HR 3-D surface structures in computer vision. Zhang et al. [54] introduced the channel attention-based spatial–spectral feature extraction network (CSSFENet) to enhance HSI feature diversity and optimize network loss functions, addressing limitations in existing HSI SR networks, particularly in feature learning challenges. Ju et al. [55] developed the super-resolution photometric stereo network (SR-PSN) to tackle the challenges of obtaining HR images in linearly responsive photometric stereo imaging systems. Zhu et al. [56] proposed Ghillie, a novel multi-illumination estimation framework based on ghost imaging theory. It utilizes a differential ghost imaging algorithm to reconstruct improved illuminance images, showcasing the effectiveness of differential algorithms in enhancing image processing performance. To integrate the self-attention mechanism with the Transformer, a novel model known as the dual aggregation transformer (DAT) [57] was introduced. This model effectively combines information across two distinct dimensions, enhancing the network's expressive capabilities. Li et al. [58] introduces the progressive split–merge super-resolution (PSMSR) framework, which incorporates a gradient-guided group attention network (GGAN) to overcome the inherent resolution limitations, which apply a multilevel split strategy from the task level, spectral level, and feature level.

The diverse approaches discussed in this section underscore the dynamic and evolving nature of HSI SR research. From innovative convolutional modules and attention mechanisms to unsupervised learning strategies, each contribution provides a unique perspective toward achieving sharper and more accurate HSI reconstructions. However, these methods continue to encounter challenges in fully exploiting the diverse features inherent in HSI. Moreover, there remains potential for improvement in the quality of reconstructed images, particularly concerning the preservation of edge details and texture quality.

III. Methodology

In this article, to introduce the proposed network, details on the differentiation and cross-domain feature extraction network (DCDFENet) for HSI super-resolution are presented in four parts: the overall framework, CDMSFE, SFDHA, EFE, and a loss function based on SVD (L_SVD). The primary objective

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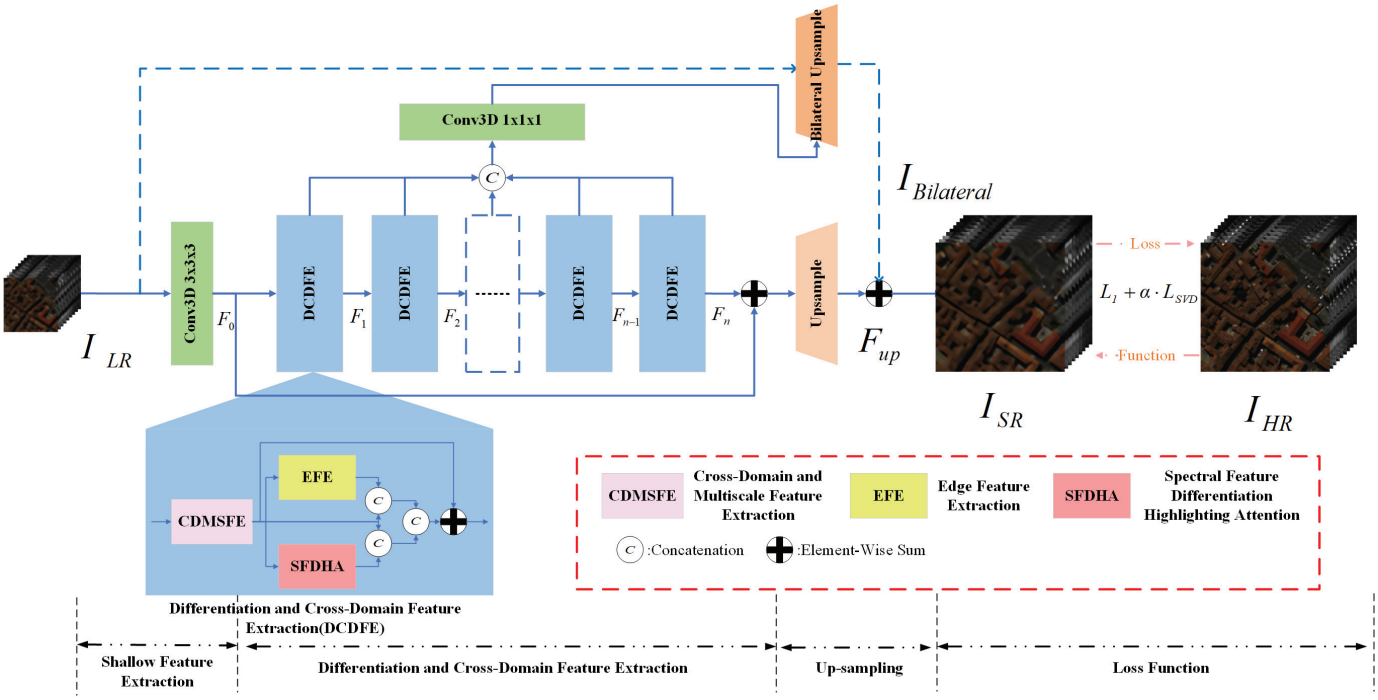


Fig. 1. Overall framework of DCDFENet.

of DCDFENet is to acquire masks for the spectral and spatial domains, respectively, using first-order and second-order differentiation masks. These masks are employed to enhance the network's feature perception capability. In addition, the network's ability to capture hyperspectral features is improved by utilizing a nonlinear mapping structure with composite receptive fields. The super-resolution process of HSIs can be expressed mathematically via the following equation:

$$I_{\mathrm{SR}} = f_\theta\left(I_{\mathrm{LR}}\right) \tag{1}$$

where I_{SR} and I_{LR} represent the SR image and LR image, θ denotes the parameters of the proposed network, and $f_\theta(\cdot)$ indicates the proposed method.

A. Overall Framework

The structure of the proposed DCDFENet network is shown in Fig. 1. It has four parts, including a shallow feature extraction module, a differentiation and cross-domain feature extraction (DCDFE) module, an image upsampling module, and a loss function. Specifically, in the first shallow feature extraction part, given the input $I_{\mathrm{LR}} \in \mathbb{R}^{B\times x\times S\times H\times W}$ (B is the batch size, S is the spectral size, and H and W are the height and width), a $3 \times 3 \times 3$ convolution filter was applied to expand the channel from 1 to C (C is set to 64) and extract the shallow feature F_0 of the input LR image I_{LR}. The feature F_0 was determined via the following equation:

$$F_0 = H_{333}\left(I_{\mathrm{LR}}\right). \tag{2}$$

Here, $H_{333}(\cdot)$ indicates the convolution operation of size $3 \times 3 \times 3$. Then, the shallow feature is sent to a sequence of n DCDFEs, which integrates differentiation features with cross-domain features to enhance the learning capability of the network

$$\begin{aligned} F_n &= H^n_{\mathrm{DCDFE}}\left(F_{n-1}\right) \\ &= H^n_{\mathrm{DCDFE}}\left(H^{n-1}_{\mathrm{DCDFE}}\left(F_{n-2}\right)\right) \\ &= H^n_{\mathrm{DCDFE}}\left(\cdots\left(H^1_{\mathrm{DCDFE}}\left(F_0\right)\right)\cdots\right). \end{aligned} \tag{3}$$

At the end of the last DCDFE, F_0 is combined with F_n through a residual shortcut and fed into the upsampling module, which is based on subpixel convolution [59], to obtain the upsampled feature $F_{\mathrm{up}} \in \mathbb{R}^{B\times 1\times S\times rH\times rW}$ (r is the upscaling factor)

$$F_{\mathrm{up}} = H_{\mathrm{upsampling}}\left(F_n + F_0\right). \tag{4}$$

To improve the transmission of features from the original I_{LR} to the end of the network, Bilateral upsampling is employed to incorporate I_{LR} and add it to F_{up}, resulting in the final SR output I_{SR}. Bilateral upsample uses the outputs of multiple DCDFE modules as the guidemap, by fully incorporating the image's grayscale information during the upsampling process, and is able to more effectively preserve the image's edge textures

$$I_{\mathrm{SR}} = I_{\mathrm{Bilateral}} + F_{\mathrm{up}} \tag{5}$$

$$I_{\mathrm{Bilateral}} = H_{\mathrm{Bilateral}}\left(I_{\mathrm{LR}}\right). \tag{6}$$

B. Cross-Domain and Multiscale Feature Extraction

HSIs contain rich features, making 3-D convolution a common choice for feature extraction. However, [60] demonstrated that combining 2-D convolution with 1-D convolution via spatial and spectral separation not only reduces the number of parameters but also significantly improves model performance compared with 3-D convolution. Therefore, in this section, a CDMSFE module is constructed on the basis of spatial and spectral separation.

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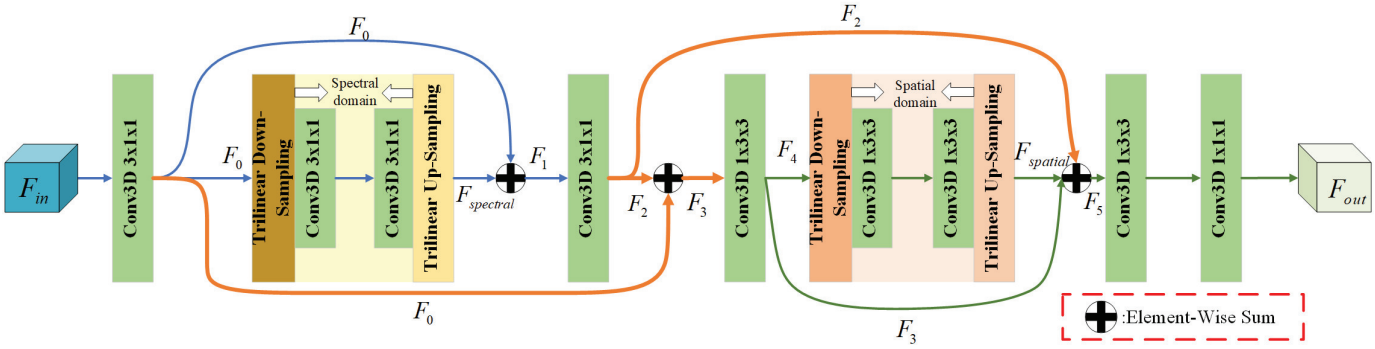


Fig. 2. Structure of CDMSFE.

As shown in Fig. 2, CDMSFE contains two main types of convolution kernels: a 2-D kernel with a size of $1 \times 3 \times 3$ and a 1-D kernel with a size of $3 \times 1 \times 1$. Using skip connections to successively concatenate these two convolutional features achieves a form of feature extraction similar to 3-D convolution. Given an input feature $F_{in} \in \mathbb{R}^{C\times S\times H\times W}$, 1-D convolution in the spectral domain with spectral downsampling is applied first. The feature F_0 is obtained through a layer of 1-D convolution in the spectral domain, followed by downsampling in the spectral dimension via trilinear sampling. Downsampling is used to reduce feature size and thereby lower computational complexity, while aggregating local information to extract more abstract features. Simultaneously, the downsampled small-scale spectral features undergo two layers of 1-D convolution in the spectral domain and are then upsampled, resulting in F_{spectral}. The feature F_1 is obtained by adding F_{spectral} and F_0. Then, F_2 is obtained by sending F_1 to an additional layer of 1-D convolution in the spectral domain

$$F_{\text{spectral}} = H_{\text{Up-tri}}\left(H_{311}\left(H_{311}\left(H_{\text{Down-tri}}\left(F_0\right)\right)\right)\right) \tag{7}$$

$$F_2 = H_{311}\left(F_1\right) = H_{311}\left(F_0 + F_{\text{spectral}}\right). \tag{8}$$

Then F_3, obtained by adding F_2 and F_0, is used as the input for 2-D convolution and spatial downsampling in the spatial domain. After 2-D convolution and spatial downsampling–upsampling, the feature F_{spatial} is obtained. The spectral-domain feature F_2, the spatial-domain feature F_{spatial}, and F_3 are subsequently combined to obtain F_3. The output feature of the module is then derived from the final feature mapping of F_3 through a 2-D convolution layer followed by a pointwise convolution layer

$$F_{\text{spatial}} = H_{\text{Up-tri}}\left(H_{133}\left(H_{133}\left(H_{\text{Down-tri}}\left(F_4\right)\right)\right)\right) \tag{9}$$

$$F_5 = F_2 + F_{\text{spatial}} + F_3 \tag{10}$$

$$F_{\text{out}} = H_{111}\left(H_{133}\left(F_5\right)\right). \tag{11}$$

F_2 and F_3 both represent features in the spectral domain. F_2 is directly fed into the spatial-domain feature extraction end for cross-domain fusion with spatial-domain features, whereas F_3 generates spatial-domain features through concatenation, resulting in cross-domain and multiscale features F_5.

C. Spectral Feature Differentiation Highlighting Attention

Given the wealth of features within the spectral dimension of HSIs, it is imperative to devise a judicious feature extraction method to leverage this spectral dimension feature information fully for HSI SR reconstruction. First-order spectral differentiation is indeed a commonly used technique in HSI analysis. By applying this method, one can effectively capture the unique features and trends present within the spectral dimensions of the HSI. The differentiation process helps to highlight important spectral features that may not be as easily discernible in the original features, making it a valuable tool for spectral analysis.

As shown in Fig. 3, we propose spectral feature. In SFDHA, differentiation and integration of spectral dimensions are employed to comprehensively extract spectral features. Concurrently, the spectral first-order differential feature map serves as a mask to facilitate attention-like weight allocation for spectral features.

Given the feature $F_{in} \in \mathbb{R}^{C\times S\times H\times W}$, differentiation is first applied along the spectral dimension, yielding $F_{\text{d-spectral}}$ as the spectral differential feature map which is shown in Fig. 3(b). Subsequently, $F_{\text{d-spectral}}$ is divided into two parallel branches. One branch utilizes convolution layers for both spectral and spatial domain to boost spectral differential features, while the other employs $\exp(-(\cdot))$ and Softmax activation functions to generate the spectral differential mask as shown in Fig. 3(c) and (d), respectively. The two features, F and F_{mask}, are then merged to obtain the enhanced feature F_{enhanced}. Subsequently, F_{enhanced} is aligned along the spectral dimension and subjected to cumulative summation to simulate integration, thereby restoring the differential feature distribution to align with that of the input features

$$F^i_{\text{d-spectral}} = F^{i+1}_{in} - F^i_{in} \tag{12}$$

$$F_{\text{mask}} = \text{Softmax}\left(\exp\left(-\left(F_{\text{d-spectral}}\right)\right)\right) \tag{13}$$

$$F_{\text{enhanced}} = F + F \cdot F_{\text{mask}} \tag{14}$$

$$F^i_{\text{out}} = H_{133}\left(H_{311}\left(\sum\nolimits_{l=0}^{i} F^l_{\text{enhanced}}\right)\right) \tag{15}$$

where F^i_{in} indicates the ith spectral band of F_{in}.

Notably, the resultant feature distribution from the differentiation operation can encompass positive or negative values. The use of the spectral differential mask is crucial for enhancing features close to the spectral extremes. Hence, special attention is warranted for eigenvalues near 0 in $F_{\text{d-spectral}}$, prompting the employment of the $\exp(-(\cdot))$ function to increase the significance of the zero point and its adjacent eigenvalues. To enhance the intuitive performance, we

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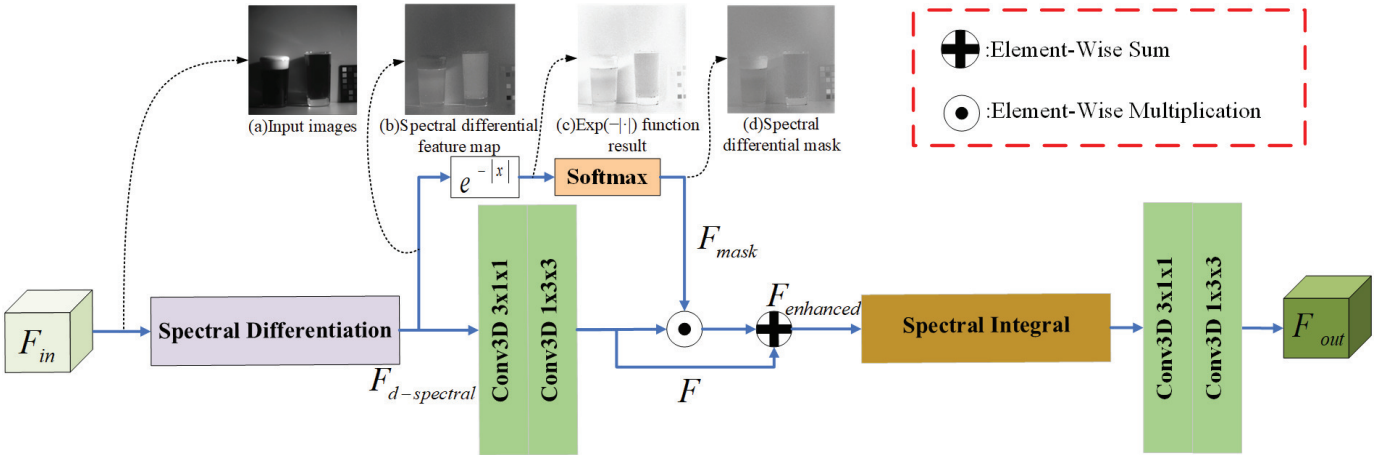


Fig. 3. Structure of SFDHA.

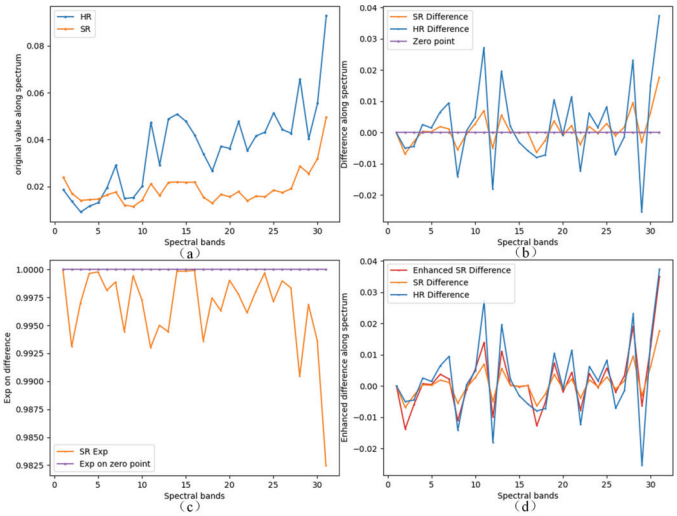


Fig. 4. (a) Contrast curves of HR and SR of SCSFINet in the spectral dimension. (b) Differential curve comparison of HR and SR of SCSFINet in the spectral dimension. (c) SR exponential function mapping curve. (d) Contrast curve of the differential curve after enhancement.

randomly sampled a series of spectral sequences from images in the CAVE dataset, magnified them 2× via SCSFINet [53], and compared them with the respective spectral sequences in the HR images. Fig. 4(a) and (b) clearly shows that there is a distinct gap in the spectral distribution between the SR and HR images generated by the SCSFINet network, both in the original data and the differential data. Fig. 4(c) demonstrates the utilization of the exponential function as a nonlinear activation function that accentuates zero points. In Fig. 4(d), the enhancement results reveal that the difference curve of the SR output aligns more closely with the HR difference curve than does the pre-enhancement state.

D. Edge Feature Extraction

Reconstructed images often suffer from blurred and missing edge details. To enhance and extract edge texture, we propose an EFE method. In this section, the Laplacian and extreme values are used to extract edge texture features in the spatial dimension. The Laplacian operator is a second-order differential operator that can be expressed as

$$\text{Laplacian} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}. \tag{16}$$

The Laplacian operator can emphasize edge information within image features. When the Laplacian is convolved in the dark region of an edge in the image, the response value is positive, whereas in the bright region of the edge, the response value becomes negative. Because pixel values in an image are typically positive, the discrepancy between the original image and the output of the Laplacian operator can amplify the edge features.

Given the high sensitivity of the Laplacian operator to noise, a module named EVFE is introduced in this section. This module is intended to utilize extreme value information from extracted image features to mitigate noise amplification caused by the Laplacian operator.

As shown in Fig. 5(a), the proposed EFE module uses the Laplacian operator to enhance the edge information of the input feature. Concurrently, EVFE is employed to extract extreme value information from the feature, thereby constraining noise amplification by the Laplacian operator

$$F_{in}^{0} = H_{133}\left(H_{311}\left(F_{in}\right)\right) \tag{17}$$

$$F_{\text{edge}}\left(x,y\right) = F_{in}^{0}\left(x,y\right) - \nabla^{2}F_{in}^{0}\left(x,y\right) \tag{18}$$

$$\nabla^{2}F_{in}^{0}\left(x,y\right) = \sum_{i,j\in[-1,1]} \text{Laplacian}\left(i,j\right)\cdot F_{in}^{0}\left(x+i,y+j\right) \tag{19}$$

where $\nabla^{2}F_{in}^{0}(x,y)$ is the second-order differential feature of $F_{in}^{0}(x,y)$ and $F_{\text{edge}}(x,y)$ represents the image after edge enhancement. The sensitivity of the Laplacian to noise primarily manifests in the following ways: when noise increases the gap between adjacent pixels, the Laplacian amplifies this gap caused by the noise. Consequently, in 2-D images, the distribution of extreme value points in the plane is disrupted. In smooth, flat regions of the image, noise can generate extreme values. Conversely, in areas characterized by significant texture changes, where pixel values vary considerably, the distribution of extreme values remains relatively stable unless the noise is exceptionally strong. Therefore, EVFE primarily utilizes the

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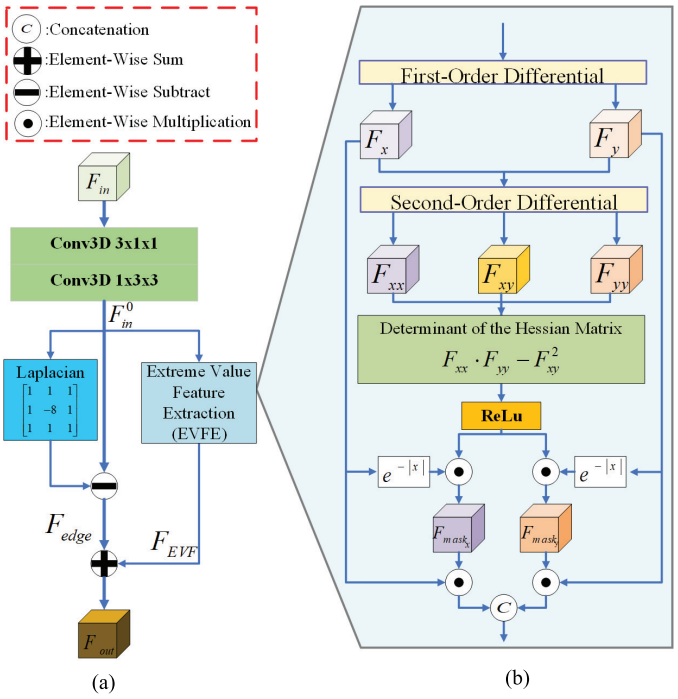


Fig. 5. Structure of EFE. (a) Structure of EFE. (b) Structure of EVFE.

extreme value distribution of input features to mitigate noise that may arise in flat regions of the image.

In Section III-C, the first-order differential in the spectral dimension signifies the change in the spectrum. Specifically, the zero-value point of the first-order differential denotes the peak and valley of the spectrum. The spatial dimension, which comprises horizontal and vertical directions, requires representation through partial differentiation. The first-order partial differential conveys the directional change trend, reflecting the texture variation in the image. Unlike the spectral differential, where the zero-value point of the first-order differential indicates the extreme value of the spectrum change, the zero-value point of the first-order partial differential in the spatial dimension may not necessarily signify extreme texture changes in that direction.

As shown in Fig. 5(b), EVFE primarily employs the mask derived from the Hessian matrix of the second-order differential to filter the extreme value from the first-order partial differential feature. The screened feature map is subsequently merged with the edge feature extracted from the first-order differential to yield the extreme value feature map.

Given the feature $F_{in}^0 \in \mathbb{R}^{C\times S\times H\times W}$, its first-order partial differential feature $F_x, F_y \in \mathbb{R}^{C\times S\times H\times W}$ can be expressed as

$$F_x(i,j) = F_{in}^0(i+1,j) - F_{in}^0(i,j) \tag{20}$$
$$F_y(i,j) = F_{in}^0(i,j+1) - F_{in}^0(i,j). \tag{21}$$

The second-order partial differential can be expressed as

$$F_{xx}(i,j) = F_x(i+1,j) - F_x(i,j) \tag{22}$$
$$F_{xy}(i,j) = F_x(i,j+1) - F_x(i,j) \tag{23}$$
$$F_{yy}(i,j) = F_y(i,j+1) - F_y(i,j). \tag{24}$$

The first-order partial differential at an extreme value of the image is 0, but a zero first-order partial differential is not necessarily indicative of an extreme value. Therefore, the Hessian matrix is also required for screening

$$F_{\text{Hessian}} = F_{xx}\cdot F_{yy} - F_{xy}\cdot F_{xy}. \tag{25}$$

When (25) is greater than 0, the position corresponding to the zero first-order partial differential is an extreme value. Conversely, when (25) is less than 0, the position corresponding to the zero first-order partial differential is a nonextreme value. Therefore, the ReLU activation function is used for F_{Hessian}, obtaining F_{mask}, which serves as a mask to filter the first-order partial differential features F_x and F_y. Notably, F_x and F_y first highlight zero positions via the function $\exp(-(\cdot))$

$$F_{\text{mask}} = \text{ReLU}(F_{\text{Hessian}}) \tag{26}$$
$$F_{\text{mask}_x} = F_{\text{mask}}\cdot\exp(-(F_x)) \tag{27}$$
$$F_{\text{mask}_y} = F_{\text{mask}}\cdot\exp\left(-\left(F_y\right)\right). \tag{28}$$

The resulting zero first-order partial differential features F_{mask_x} and F_{mask_y} are subsequently combined with the original first-order partial differential features F_x and F_y to obtain the final extreme value feature F_{EVF}

$$F_{\text{EVF}} = H_{111}\left(\text{cat}\left(F_{\text{mask}_x}, F_{\text{mask}_y}\right)\right) \tag{29}$$

where $H_{111}(\cdot)$ is the convolution kernel with a size of $1\times 1\times 1$. Finally, the final output feature F_{out} of EVF is obtained by summing F_{edge} and F_{EVF}

$$F_{\text{out}} = F_{\text{edge}} + F_{\text{EVF}}. \tag{30}$$

E. Loss Function Based on SVD (L_SVD)

SVD is a matrix factorization method similar to eigenvalue decomposition, but while eigenvalue decomposition is limited to square matrices, SVD can be applied to any matrix. By decomposing an image matrix, the singular value set of the image can be obtained and sorted in the descending order. The singular vectors corresponding to the largest singular values represent important feature vectors in the original matrix. By selecting the singular values, the main components of the matrix can be extracted from the original matrix.

Given a nonzero real matrix $A \in \mathbb{R}^{m\times n}$, m and n are the height and width of matrix $A \in \mathbb{R}^{m\times n}$, respectively. A can be represented as the product of three operations in the form of real matrices, which is a type of matrix factorization

$$A = U\Sigma V^T. \tag{31}$$

Here, U is an orthonormal square matrix of order m, V is an orthonormal square matrix of order n, and Σ is a diagonal matrix composed of nonnegative diagonal elements in descending order, representing the singular values. The singular value satisfies the following:

$$UU^T = E \tag{32}$$
$$\Sigma = \text{diag}(\sigma_1, \sigma_2, \ldots, \sigma_n) \tag{33}$$
$$p = \min(m,n) \tag{34}$$
$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0. \tag{35}$$

Here, E is the identity matrix and σ_i represents the ith singular value of matrix A, where $i = 0, 1, \ldots, p$.

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Every nonzero real matrix possesses an SVD. When an image is represented as a 2-D matrix, the vectors corresponding to the singular values of the image denote the feature set contained within it. For HSIs, SVD can be applied to each spectral band to obtain singular values that indicate the significance of each spectrum in the spectral dimension. In HSIs, a response curve is distributed along the spectral dimension, demonstrating that different spectral bands exhibit varying weight ratios across the entire spectral range. Consequently, this section proposes a loss function based on SVD (L_{SVD}). From the perspective of spectral singular values, the HR image and the reconstructed SR image are decomposed via SVD to obtain their respective singular value sets

$$L_{\mathrm{SVD}} = \frac{1}{p}\sum_{i=1}^{p} |\mathrm{SVD}(I_{\mathrm{SR}}) - \mathrm{SVD}(I_{\mathrm{HR}})|. \tag{36}$$

Here, I_{SR} is the reconstructed SR image and I_{HR} is the corresponding HR image.

L_1 norm as the loss function combined with the L_{SVD} loss function is used to constrain the network from the perspectives of both the image structure and the singular value

$$L_1 = \frac{1}{N}\sum_{i=1}^{N} \left| I_{\mathrm{SR}}^i - I_{\mathrm{HR}}^i \right| \tag{37}$$

where N represents the number of training samples.

Ultimately, our loss function is expressed as follows:

$$L_{\mathrm{total}} = L_1 + \alpha L_{\mathrm{SVD}} \tag{38}$$

where α represents the weight of the L_{SVD} and our experimental results indicate that the optimal performance is achieved when α is set to 0.01.

IV. Experiments

In this section, a comprehensive evaluation of the proposed DCDFENet is presented, examining both quantitative and qualitative aspects. Initially, the DCDFENet is tested using three widely utilized datasets, with specific implementation details provided. Subsequently, an in-depth performance analysis of DCDFENet is conducted. Finally, a comparison is made between the proposed DCDFENet and other methods, including Bicubic, VDSR [61], EDSR [62], MCNet [43], ERCSR [44], MSFMNet [49], SCSFINet [53], and SRDNet [63].

A. Datasets

Among the numerous HSI datasets, we select three most common datasets as benchmarks to verify the performance of the proposed network. These datasets are CAVE, Pavia Center, and Pavia University.

1) CAVE: The CAVE dataset was collected via a cooled CCD camera with a spectral range of 400–700 nm. Each image in the dataset has a resolution of pixels and contains gray-level information across 31 spectral bands, with each band spaced 10 nm apart. The dataset includes a total of 32 HSIs. During the training phase, we selected 25 images for training and reserved seven images for testing.

2) Pavia Center (Pavia) and Pavia University (PaviaU): The Pavia dataset consists of 102 spectral bands, whereas the PaviaU dataset consists of 103 spectral bands. Both datasets were acquired via a ROSIS-03 sensor with a spectral range of 0.43–0.86 μm and a spatial resolution of 1.3 m. To facilitate more efficient training and testing, we randomly selected a pixel patch from both Pavia and PaviaU datasets as the test images and used the remaining regions as the training images.

B. Evaluation Metrics

The task of super-resolution reconstruction requires objective evaluation indicators to assess the effectiveness of the reconstruction algorithms. Therefore, four evaluation methods—the peak signal-to-noise ratio (PSNR), mean PSNR (MPSNR), structural similarity (SSIM), and spectral angle mapping (SAM)—were selected to quantitatively evaluate the performance of the network reconstruction results. These methods are defined as follows:

$$\mathrm{mse} = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W}(X(i,j) - Y(i,j))^2 \tag{39}$$

$$\mathrm{PSNR} = 10\times \log_{10}\frac{\left(2^{\mathrm{Bits}}-1\right)^2}{\mathrm{mse}} \tag{40}$$

$$\mathrm{MPSNR} = \frac{10}{B}\sum_{i=1}^{B}\log_{10}\frac{\left(2^{\mathrm{Bits}}-1\right)^2}{\mathrm{mse}} \tag{41}$$

$$\mathrm{SSIM} = \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\sigma_{xy} + C_2\right)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)} \tag{42}$$

$$\theta(z^*, z_h) = \cos^{-1}\left(\frac{z_h^T z^*}{\sqrt[2]{(z^*)^T z^*}\sqrt[2]{z_h^T z_h}}\right) \tag{43}$$

$$\mathrm{SAM} = \frac{1}{H\cdot W}\sum_{i=1}^{H}\sum_{j=1}^{W}\theta(X(i,j), Y(i,j)) \tag{44}$$

where mse$(\cdot)$ denotes the mean square error function. X and Y denote the reconstructed HSI I_{SR} and the original ground-truth image I_{HR}, respectively. i and j indicate the index of the image height H and width W, respectively. Bits indicates the image pixel depth and B indicates the number of spectral bands of the HSI image. μ_x and μ_y indicate the mean value of I_{SR} and I_{HR}. σ_x and σ_y denote the variance of I_{SR} and I_{HR}, respectively. σ_{xy} indicates the covariance between I_{SR} and I_{HR} while $C1$ and $C2$ indicate the minima that prevent division by 0. $\theta(\cdot)$ represents the SAM function. z^* and z_h represent the pixel vector of I_{SR} and I_{HR}, respectively, and T represents the vector transpose operation.

C. Implementation Details

The data preparation, network training, and testing described in this study were conducted in the MATLAB, Python 3.8, and PyTorch 1.10 environments. The hardware device consisted of four NVIDIA RTX3090 graphics cards. In the training phase, we optimize the model using the Adam optimizer, whose parameters are set $\beta_1 = 0.9$, $\beta_2 = 0.999$. The initial learning rate is set to $1e^{-3}$ and the total epochs are set to 400.

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TABLE I
QUANTITATIVE COMPARISONS OF HSI SUPER-RESOLUTION ON THE CAVE DATASET

Scale	Metric	Bicubic	VDSR	EDSR	MCNet	ERCSR	MSFMNet	SCSFINet	SRDNet	Ours
×2	PSNR↑	40.330	44.456	45.151	45.878	45.972	46.015	46.240	45.618	**46.667**
	MPSNR↑	39.500	43.531	44.207	44.913	45.038	45.039	45.240	44.599	**45.728**
	SSIM↑	0.9820	0.9895	0.9907	0.9913	0.9914	0.9917	0.9921	0.9913	**0.9922**
	SAM↓	3.311	2.866	2.606	2.588	2.544	2.497	**2.474**	2.591	2.527
×4	PSNR↑	34.616	37.027	38.117	38.589	38.626	38.733	38.848	38.476	**39.395**
	MPSNR↑	33.657	36.045	37.137	37.679	37.738	37.814	37.897	37.488	**38.471**
	SSIM↑	0.9388	0.9591	0.9626	0.9690	0.9695	0.9697	0.9699	0.9670	**0.9725**
	SAM↓	4.784	4.297	4.132	3.682	3.643	3.676	3.630	3.813	**3.546**
×8	PSNR↑	30.554	32.184	33.416	33.607	33.624	33.675	33.723	33.623	**34.054**
	MPSNR↑	29.484	31.210	32.337	32.520	32.556	32.599	32.638	32.489	**32.996**
	SSIM↑	0.8657	0.8852	0.9002	0.9125	0.9113	0.9136	0.9134	0.9071	**0.9162**
	SAM↓	6.431	5.747	5.409	5.172	5.114	5.084	5.027	5.329	**4.962**

D. Comparison and Analysis

In this section, we compared our method with seven state-of-the-art methods, including Bicubic, VDSR [61], EDSR [62], MCNet [43], ERCSR [44], MSFMNet [49], SCSFINet [53], and SRDNet [63] on three datasets at different scale factors: 2, 4, and 8. VDSR and EDSR are advanced RGB SR algorithms based on 2-D convolution. ERCSR introduces an alternating structure of 2-D and 3-D convolution units to enhance spatial learning and reduce structural redundancy. MCNet adopts a novel mixed convolutional module to extract the potential features by 2-D/3-D convolution, which enables the network to more mine spatial features. MSFMNet focuses on effectively extracting and integrating multiscale features from HSIs to improve their spatial resolution. SCSFINet is an HSI SR network based on spectrum-guided attention for analyzing the information already obtained from HSIs. SRDNet adopts a dual-domain network with hybrid 2-D/3-D convolutions and progressive upsampling to jointly exploit spatial and spectral information. In addition, a frequency-domain loss is introduced to enhance high-frequency details.

1) Results on the CAVE Dataset: Table I summarizes the average quantitative super-resolution performance of the proposed framework and eight other methods on the CAVE dataset. The best result is highlighted in bold, and the second-best result is indicated by an underline. The framework demonstrates superior performance on most evaluation metrics, including PSNR, MPSNR, SSIM, and SAM. Specifically, improvements of 0.427, 0.547, and 0.331 dB in PSNR are observed at the ×2, ×4, and ×8 scales, respectively, compared with the second methods. In the context of the MPSNR metric, DCDFENet shows enhancements of 0.488, 0.574, and 0.358 dB over the second-best results at the ×2, ×4, and ×8 scales, respectively. For the SSIM metric, the improvements relative to the second-best outcomes are recorded as 0.0001, 0.0026, and 0.0026. Moreover, promising results are achieved in the SAM metric, with enhancements of 0.084 and 0.065 at the ×4 and ×8 scales, respectively. This performance is attributed to the framework's utilization of cross-domain multiscale composite receptive fields for feature extraction, combined with spectral differencing to effectively harness spectral information. Furthermore, the application of Laplacian operations and extremum feature enhancement considerably improved the network's perception of image edge information. The bicubic method exhibited the poorest performance, primarily due to its simple algorithm structure, which is inadequate for capturing sufficient spatial–spectral information. Furthermore, the results generated by VDSR and EDSR methods were notably fuzzy and exhibited significant texture changes, attributed to the improper use of spectral and spatial information in HSIs. While these methods demonstrate improvements in spatial reconstruction accuracy, as indicated by PSNR and SSIM, they are deficient in spectral fidelity (SAM). MCNet, which employs a hybrid convolution approach, effectively exploits spectral correlations while reconstructing spatial resolution. However, it relies exclusively on spatial loss to constrain network learning, lacking frequency domain modeling, which results in suboptimal performance. Although MSFMNet and ERCSR have developed fundamental modules specifically for hyperspectral datasets, these modules depend on basic 2-D/3-D convolutions to extract 2-D and 3-D features without leveraging the interconnections between the full spectral bands and spatial information. SCSFINet effectively exploits the coupling relationship between the spectral and spatial domains and conducts multilevel fusion but exhibits limitations in the reconstruction of image edge textures. SRDNet employs a 2-D/3-D hybrid convolution module for joint spatial–spectral feature extraction, along with a hierarchical self-attention mechanism to capture long-range spatial dependencies and spectral correlations. However, experimental results indicate that the network still shows limitations in recovering fine texture details.

Fig. 6 presents the visual results of the reconstructed fake_and_real_lemons_ms image from the CAVE dataset with a scale factor of 8 and the normalized error heatmap. The 26th, 17th, and 9th spectral bands of the reconstructed result are used as the RGB channel with a scale factor of 8. The bluer the error map, the better the reconstruction performance. We selected representative areas and marked them with red boxes. From the subjective comparison images, it can be observed that the images reconstructed by our framework structure have the best visual effects, particularly in terms of texture and edge details. Compared with the results of other comparative

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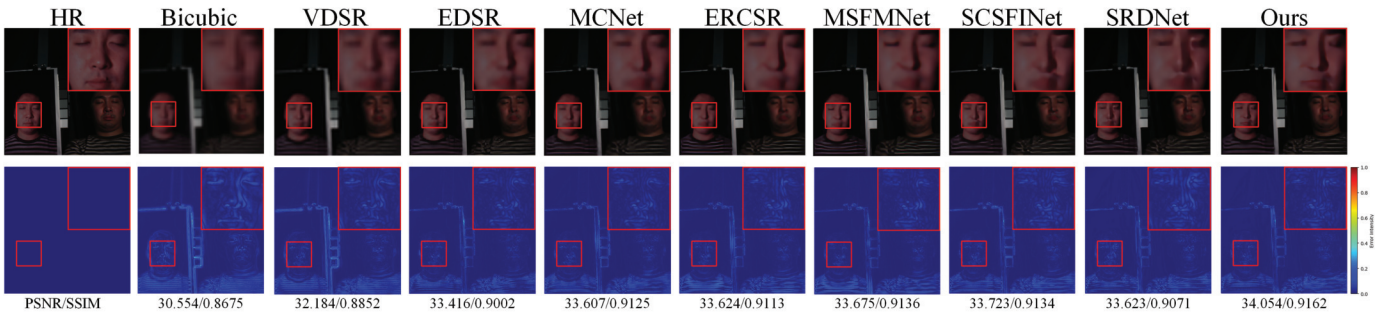


Fig. 6. Visual comparisons of the reconstructed photo_and_face_ms image on the CAVE dataset at ×8 scale.

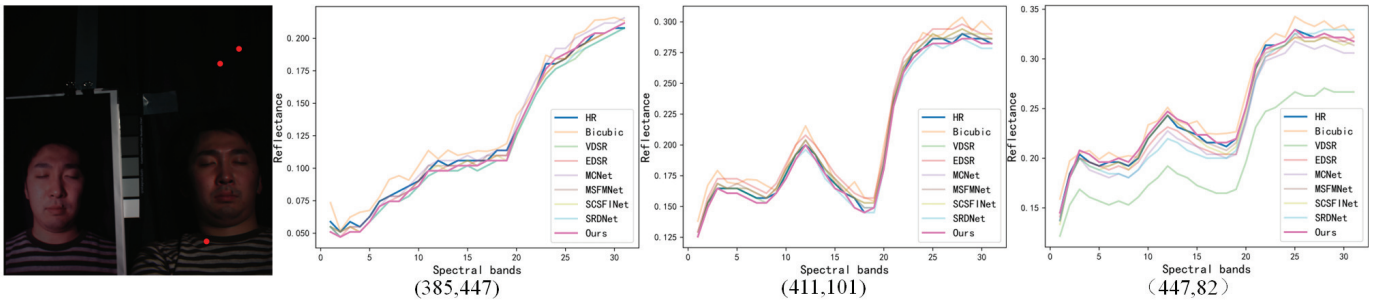


Fig. 7. Visual comparison of the photo_and_face_ms image spectral curves at pixel locations (385, 447), (411, 101), and (447, 82) on the CAVE dataset.

TABLE II
QUANTITATIVE COMPARISONS OF HSI SUPER-RESOLUTION ON THE PAVIA DATASET

Scale	Metric	Bicubic	VDSR	EDSR	MCNet	ERCSR	MSFMNet	SCSFINet	SRDNet	Ours
×2	PSNR↑	32.406	35.392	35.160	35.124	35.602	35.678	35.927	35.422	**36.075**
	MPSNR↑	31.798	34.879	34.580	34.626	35.099	35.200	35.413	34.848	**35.503**
	SSIM↑	0.9036	0.9501	0.9452	0.9455	0.9503	0.9560	0.9540	0.9487	**0.9563**
	SAM↓	4.370	3.689	3.898	3.865	3.683	3.656	3.627	3.701	**3.531**
×4	PSNR↑	26.596	28.328	28.649	28.791	28.862	28.920	29.031	29.211	**29.557**
	MPSNR↑	26.556	28.317	28.591	28.756	28.815	28.873	29.000	29.148	**29.467**
	SSIM↑	0.7091	0.7707	0.7782	0.7826	0.7818	0.7863	0.7943	0.8019	**0.8133**
	SAM↓	7.553	6.514	6.573	6.385	6.125	6.300	5.873	**5.636**	5.644
×8	PSNR↑	24.464	24.526	24.854	24.877	24.965	25.027	25.125	25.130	**25.306**
	MPSNR↑	24.745	24.804	25.067	25.096	25.190	25.257	25.377	25.346	**25.483**
	SSIM↑	0.4899	0.4944	0.5282	0.5391	0.5382	0.5464	0.5553	0.5547	**0.5717**
	SAM↓	7.648	7.588	7.507	7.429	7.834	7.449	7.404	**7.289**	7.609

algorithms, the facial features within red boxes are clearer and closer to HR images.

The reflectance curves of the pixels randomly selected in different spectral bands in the results obtained via different methods are shown in Fig. 7. The curve of DCDFENet was closer to the original HR image curve, which can recover the spectral features of the target well.

2) Results on the Pavia Dataset: Table II presents the average quantitative super-resolution performance of the proposed framework alongside eight other methods on the Pavia dataset. The best results are highlighted in bold, while the second-best results are indicated by an underline. The framework demonstrates superior performance on most evaluation metrics. Specifically, improvements of 0.148, 0.346, and 0.176 dB in PSNR are observed at ×2, ×4, and ×8 scales, respectively, compared with the second-best methods. In the context of the MPSNR metric, DCDFENet shows enhancements of 0.090, 0.319, and 0.106 dB over the second-best results at ×2, ×4, and ×8 scales, respectively. For the SSIM metric, the improvements relative to the second-best outcomes are recorded as 0.0003, 0.0114, and 0.0164. Moreover, promising results are achieved in the SAM metric, with enhancements of 0.096 at the ×2 scale. Benefiting from the explicit modeling of spectral frequency-domain loss, SRDNet achieves slightly better performance in terms of SAM at ×4 and ×8. However, our method performs favorably in terms of PSNR and SSIM, indicating its superior ability to capture spatial structures.

Fig. 8 displays the visualization results of the reconstructed images for a scale factor of 4, along with the normalized error heatmap. To enhance the visualization of HSI, the 64th, 35th, and 13th spectral bands are selected as RGB channels to generate pseudo-color RGB images. The PSNR and SSIM of the corresponding HSI for each method are reported at the bottom. An examination of the region delineated by the

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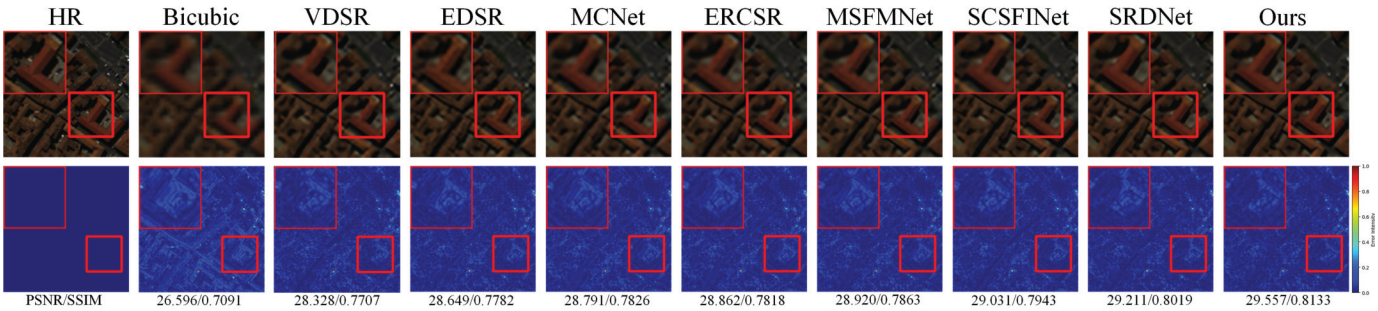


Fig. 8. Visual comparisons of the reconstructed valid image on the Pavia dataset at $\times 4$ scale.

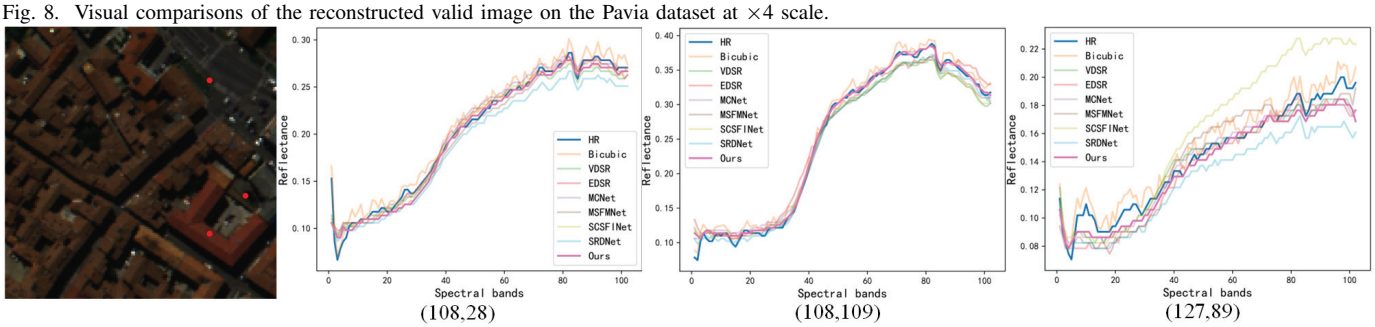


Fig. 9. Visual comparison of the reconstructed valid image spectral curves at pixel locations (108, 28), (108, 109), and (127, 89) on the Pavia dataset.

TABLE III

QUANTITATIVE COMPARISONS OF HSI SUPER-RESOLUTION ON THE PAVIAU DATASET

Scale	Metric	Bicubic	VDSR	EDSR	MCNet	ERCSR	MSFMNet	SCSFINet	SRDNet	Ours
$\times 2$	PSNR↑	30.509	33.988	33.943	33.695	33.857	34.807	35.914	34.923	**36.122**
	MPSNR↑	30.497	34.038	33.985	33.743	33.910	34.98	35.033	35.019	**36.316**
	SSIM↑	0.9055	0.9524	0.9511	0.9502	0.9520	0.9582	0.9584	0.9591	**0.9650**
	SAM↓	3.816	3.258	3.334	3.359	3.220	3.16	3.068	2.979	**2.866**
$\times 4$	PSNR↑	29.061	29.761	29.795	29.889	30.049	30.14	30.388	30.380	**30.900**
	MPSNR↑	29.197	29.904	29.894	29.993	30.164	30.283	30.489	30.477	**30.949**
	SSIM↑	0.7322	0.7753	0.7791	0.7835	0.7899	0.7948	0.8068	0.8076	**0.8295**
	SAM↓	5.248	4.997	5.074	4.917	4.865	4.861	4.692	4.577	**4.511**
$\times 8$	PSNR↑	26.699	26.737	27.182	27.201	27.288	27.334	27.384	27.508	**27.766**
	MPSNR↑	26.99	27.028	27.467	27.483	27.548	27.586	27.590	27.729	**27.972**
	SSIM↑	0.5936	0.5962	0.6302	0.6254	0.6276	0.6356	0.6427	0.6547	**0.6831**
	SAM↓	7.179	7.133	6.678	6.683	6.611	6.615	6.576	6.208	**6.057**

red box reveals that the image reconstructed by the framework displays improved clarity in the edge details of the buildings and a richer texture in areas of internal variation. This reconstruction aligns more closely with the HR reference, highlighting the superior performance of the network in reconstructing edge textures compared with other algorithms. The comparative results of the error plots indicate that the reconstruction outcome of the network exhibits the darkest coloration, corresponding to the lowest error magnitude, signifying the most accurate reconstruction among the evaluated methods.

Fig. 9 illustrates the spectral curves of randomly selected pixels on the reconstructed images from the Pavia dataset, with an image scale factor of 4, demonstrating that the spectral curve of DCDFENet is closer to the original HR image curve, thereby achieving higher spectral fidelity compared with other methods.

3) Results on the PaviaU Dataset: Table III presents the quantitative descriptions of the four performance indicators for scale factors $\times 2$, $\times 4$, and $\times 8$ in the PaviaU dataset. The experimental results clearly demonstrate the superiority of our proposed method over other existing methods in analyzing PaviaU datasets at different scales. Considering that both Pavia and PaviaU datasets were smaller than the CAVE dataset, we found that our method also performed well with a small dataset. In terms of the overall numerical performance, the trends presented are basically consistent with those of the Pavia dataset. Our method outperforms the second-best method in terms of all metrics on all three scales. With respect to the PSNR metric, there are improvements of 0.208, 0.512, and 0.258 dB for $\times 2$, $\times 4$, and $\times 8$ scale factors, respectively. Meanwhile, the MPSNR metric shows significant enhancements of 1.283, 0.460, and 0.243 dB across the three scaling factors. The SSIM also improves by 0.0059, 0.0219, and

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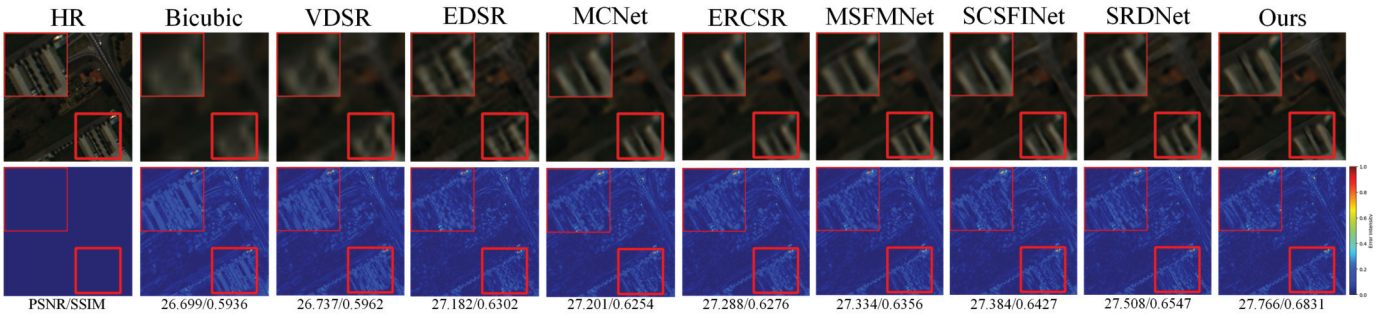


Fig. 10. Visual comparisons of the reconstructed valid image on the PaviaU dataset at $\times 8$ scale.

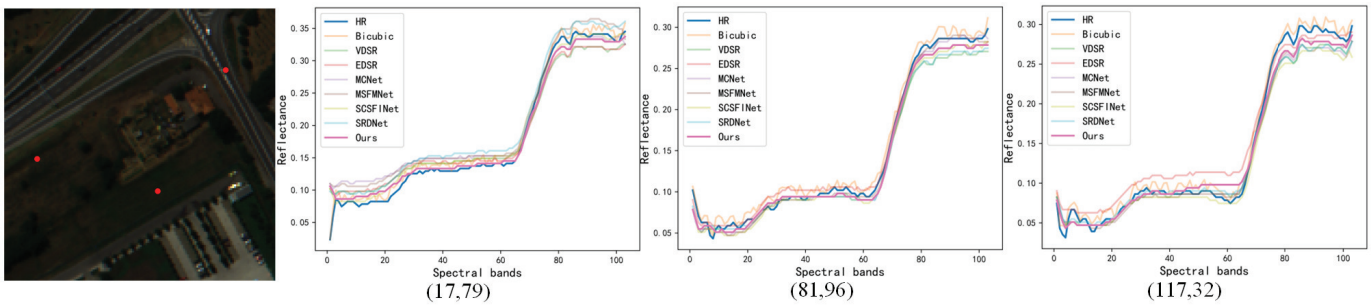


Fig. 11. Visual comparison of the reconstructed valid image spectral curves at pixel locations (17, 79), (81, 96), and (117, 32) on the PaviaU dataset.

TABLE IV
COMPARISONS OF PARAMETER NUMBER AND FLOPS(G) UNDER DIFFERENT UPSCALING FACTORS ON THE PAVIAU DATASET

Method	Scale×2		Scale×4		Scale×8	
	Param.	FLOPs	Param.	FLOPs	Param.	FLOPs
Bicubic	-	-	-	-	-	-
VDSR	0.82M	5.17G	0.97M	2.98G	1.12M	8.61G
EDSR	1.49M	8.61G	1.63M	3.84G	1.78M	2.65G
MCNet	1.93M	1343.10G	2.17M	1178.38G	2.96M	2685.64G
ERCSR	1.35M	1092.07G	1.59M	1172.21G	2.38M	2740.69G
MSFMNet	2.75M	1720.37G	2.75M	430.19G	2.75M	107.65G
SCSFINet	4.16M	1843.45G	4.16M	460.97G	4.17M	115.35G
SRDNet	1.70M	719.57G	1.73M	182.97G	1.81M	50.37G
Ours	2.52M	1245.11G	2.52M	311.48G	2.53M	78.08G

0.0284, respectively. Finally, with respect to the SAM metric, the improvements are 0.113, 0.066, and 0.151, respectively.

Table IV compares the computational complexity of various competing methods at different upscaling factors on the PaviaU dataset, in terms of the number of parameters and FLOPs. As shown in the table, our proposed method achieves a favorable balance between reconstruction performance and computational cost.

Specifically, our method maintains a moderate parameter count across all scales, which is significantly lower than that of SCSFINet and MSFMNet, and comparable to SRDNet. Regarding FLOPs, although our approach incurs slightly higher computational cost than SRDNet, it remains considerably more efficient than MCNet, ERCSR, MSFMNet, and SCSFINet, whose computational costs exceed 1000G at certain scales. Importantly, our model maintains high reconstruction quality while offering substantial computational savings. Although SRDNet achieves relatively lower FLOPs, its overall reconstruction quality is inferior, indicating a tradeoff between computational efficiency and output fidelity. In contrast, our method maintains a better balance between complexity and performance and demonstrates enhanced capability in reconstructing fine spatial details.

Fig. 10 shows the visualization results of the reconstructed images in the $\times 4$ SR as well as the normalized error heatmap. We select bands 64, 35, and 13 for pseudo-color image synthesis. The image reconstructed using our method has a significantly richer texture. In addition, the road reconstructed via our method is slenderer and has a weaker fuzzy effect than those of the other methods. In the mean spectral difference curve shown in Fig. 11, our method also has obvious advantages.

E. Ablation Study

We evaluated the proposed network from four aspects, including CDMSFE, SFDHA, and EFE. To compare the modules, we determined the results for a scale factor of 4 using PaviaU and CAVE datasets.

1) Study of CDMSFE: To better test the effectiveness of the CDMSFE, we designed three ablation experiments: CDMSFE-a, CDMSFE-b, and CDMSFE-c. For the CDMSFE-a experiment, we reduced the composition of the CDMSFE module to four simple concatenated standard 3-D convolutions. For the CDMSFE-b experiment, we eliminated the spectral and spatial-domain sampling operations from CDMSFE, retaining only the corresponding convolution operations. For the CDMSFE-c experiment, we removed the skip connections in CDMSFE related to cross domains.

Table V presents the metrics of three ablation structures at the upsampling factor of 4 on the PaviaU and CAVE datasets. All three ablation structures demonstrate varying degrees of performance degradation compared with the original framework in all metrics.

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TABLE V

ABLATION STUDY RESULTS FOR EVALUATING THE EFFICIENCY OF CDMSFE MODULES ON PAVIAU AND CAVE DATASETS WITH A SCALE FACTOR OF 4

Model	#Params	Dataset	PSNR↑	MPSNR↑	SSIM↑	SAM↓
Standard	2.52M	PaviaU	**30.900**	**30.949**	**0.8295**	**4.511**
		CAVE	**39.395**	**38.471**	**0.9725**	**3.546**
CDMSFE-a	2.03M	PaviaU	30.702	30.761	0.8231	4.599
		CAVE	38.990	38.091	0.9711	3.610
CDMSFE-b	2.52M	PaviaU	30.507	30.566	0.8133	4.597
		CAVE	39.027	38.109	0.9707	3.651
CDMSFE-c	2.46M	PaviaU	30.762	30.865	0.8223	4.550
		CAVE	38.838	37.925	0.9697	3.836

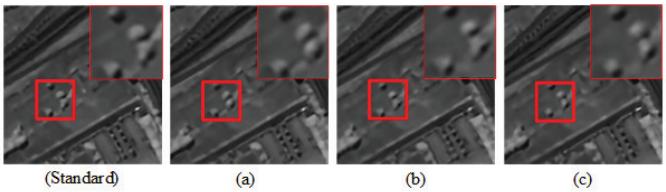


Fig. 12. Visual comparisons of the 85th spectral band of reconstructed images from different ablation experiments of CDMSFE on PaviaU dataset. (a) CDMSFE-a. (b) CDMSFE-b. (c) CDMSFE-c.

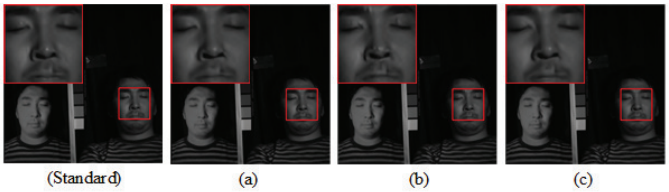


Fig. 13. Visual comparisons of the 28th spectral band of reconstructed images from different ablation experiments of CDMSFE on CAVE dataset. (a) CDMSFE-a. (b) CDMSFE-b. (c) CDMSFE-c.

Specifically, replacing the module with a simple 3-D convolution in the CDMSFE-a model leads to a decrease in parameter count. However, performance also deteriorates. This finding indicates the effectiveness of the cross-domain connections and the upsampling–downsampling structures in the spectral and spatial domains for extracting and utilizing spatial–spectral correlation information. The performance of Structure B further degrades when the upsampling–downsampling operations are removed, as this impairs the network's ability to represent multiscale features. Structure C, which lacks cross-domain connections, results in limited interaction between features from different domains, thereby affecting the network's feature extraction capabilities.

The visual comparison of reconstruction results from different ablation structures is shown in Figs. 12 and 13. It can be observed that the absence of modules in various structures leads to varying degrees of detail loss in the reconstructed images, further demonstrating the effectiveness of our proposed modules.

2) Study of SFDHA: To better test the effectiveness of SFDHA, we designed two additional experiments: SFDHA-a and SFDHA-b. For SFDHA-a, we retained only the convolutional operation part, meaning that SFDHA-a consists of four convolutional layers in series. For SFDHA-b, we kept the differential and integral operations in the spectral domain, along with the convolution operations.

TABLE VI

ABLATION STUDY RESULTS FOR EVALUATING THE EFFICIENCY OF SFDHA MODULES ON PAVIAU AND CAVE DATASETS WITH A SCALE FACTOR OF 4

Model	#Params	Dataset	PSNR↑	MPSNR↑	SSIM↑	SAM↓
Standard	2.52M	PaviaU	**30.900**	**30.949**	**0.8295**	**4.511**
		CAVE	**39.395**	**38.471**	**0.9725**	**3.546**
SFDHA-a	2.46M	PaviaU	30.581	30.727	0.8151	4.794
		CAVE	38.974	38.059	0.9718	3.640
SFDHA-b	2.48M	PaviaU	30.822	30.957	0.8291	4.572
		CAVE	39.236	38.323	0.9727	3.642

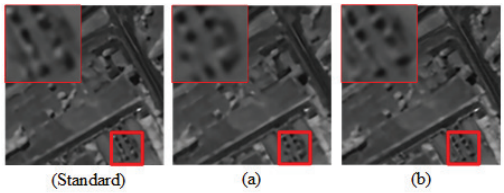


Fig. 14. Visual comparisons of the 85th spectral band of reconstructed images from different ablation experiments of SFDHA on PaviaU dataset. (a) SFDHA-a. (b) SFDHA-b.

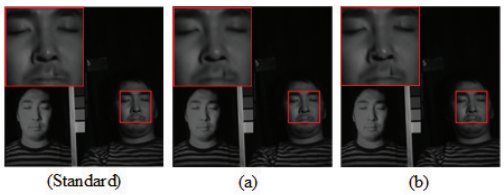


Fig. 15. Visual comparisons of the 28th spectral band of reconstructed images from different ablation experiments of SFDHA on CAVE dataset. (a) SFDHA-a. (b) SFDHA-b.

Table VI presents the metrics of three ablation structures at the upsampling factor of 4 on PaviaU and CAVE datasets. When the internal structure is simplified to only four convolutional layers in the SFDHA-a model, a significant decline in performance is observed, adversely affecting both spatial resolution reconstruction and spectral reconstruction abilities. In Structure SFDHA-b, where spatial-domain differentiation and integration operations are retained on top of the convolutional layers (with the other auxiliary branch removed), the trend of performance decline is mitigated. This finding demonstrates that the spectral differentiation operation effectively highlights regions with significant spectral changes, enabling the network to better extract global spectral information. The auxiliary branch processes the results of spectral differentiation through an attention mechanism, creating a mask that enhances focus on key regions requiring attention. Experiments further confirm the effectiveness of this auxiliary branch.

The visual comparison of reconstruction results from dif ferent ablation structures is shown in Figs. 14 and 15.

3) Study of EFE: To better test the effectiveness of EFE, we designed two additional experiments: EFE-a and EFE-b. For EFE-a, we removed both the Laplacian and the EVFE parts. For EFE-b, we removed only the EVFE part. Table VII presents the metrics of three ablation structures at the upsampling factor of 4 on the PaviaU and CAVE datasets. The data indicate that both ablation frameworks for EFE exhibit a certain degree of performance decline compared with the

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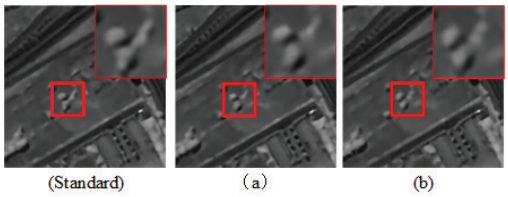


Fig. 16. Visual comparisons of the 85th spectral band of reconstructed images from different ablation experiments of EFE on PaviaU dataset. (a) EFE-a. (b) EFE-b.

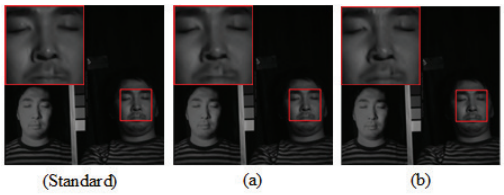


Fig. 17. Visual comparisons of the 28th spectral band of reconstructed images from different ablation experiments of EFE on CAVE dataset. (a) EFE-a. (b) EFE-b.

TABLE VII

ABLATION STUDY RESULTS FOR EVALUATING THE EFFICIENCY OF EFE MODULES ON PAVIAU AND CAVE DATASETS WITH A SCALE FACTOR OF 4

Model	#Params	Dataset	PSNR↑	MPSNR↑	SSIM↑	SAM↓
Standard	2.52M	PaviaU	**30.900**	**30.949**	**0.8295**	**4.511**
		CAVE	**39.395**	**38.471**	**0.9725**	**3.546**
EFE-a	2.15M	PaviaU	30.777	30.861	0.8206	4.513
		CAVE	38.907	37.966	0.9700	3.791
EFE-b	2.48M	PaviaU	30.811	30.943	0.8254	4.538
		CAVE	39.081	38.142	0.9713	3.619

standard network. The structure EFE-a, which removes the entire main framework, shows the most significant decrease in performance, highlighting the effectiveness of our original structure. In Experiment B, we retained the Laplacian operation to enhance the extraction of image edge information while removing the EVFE part. The performance showed some recovery, demonstrating the auxiliary role of EVFE in edge information extraction. The visual comparison of reconstruction results from different ablation structures of EFE is shown in Figs. 16 and 17.

V. CONCLUSION

In this article, we propose DCDFENet for HSI SR. The core of the network is the DCDFE module, which comprises three primary components. First, the CDMSFE module establishes complementary relationships between spectral and spatial domains via cross-domain interactions with spatial and spectral separation 3-D convolutions. Second, the SFDHA module captures variation trends along the spectral dimension through spectral differencing and employs an attention mechanism to adaptively regulate the influence of spectral variations, thereby facilitating effective interspectral feature extraction. Third, the EFE module enhances spatial edge and texture representations by integrating a Laplacian operator with an EVFE submodule. In addition, a SVD-based loss function (L_SVD) is introduced to alleviate spectral distortion. Extensive quantitative and qualitative experiments conducted on multiple hyperspectral datasets demonstrate the effectiveness and generalization capability of the proposed method, achieving superior performance in both objective metrics and visual quality assessments.

Although the proposed network has demonstrated outstanding performance across various evaluation metrics, there is still room for improvement in spectral-domain reconstruction. In addition, optimizing the computational complexity while preserving high reconstruction accuracy remains a critical issue. Future work will focus on enhancing the spectral fidelity of reconstructed images and developing more lightweight and efficient network designs to further improve the applicability of the model in real-world remote sensing scenarios.

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